

Results 2 - A Triangular Tank

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1 Results and Discussion

1.1 A Triangular Tank

We will now calculate the sloshing modes and frequencies of fluid in a two-dimensional triangular tank, where the side walls subtend an angle of 45 degrees with the free surface. This problem does indeed admit an analytical solution and the reader is directed to Lamb[2] or the original paper by Kirchhoff[1] for a complete derivation. The analytic streamfunction and velocity potential for both the symmetric and asymmetric modes are listed below (omitting the time factor) for a triangular tank of height h with the undisturbed free surface along the x-axis:

The symmetric modes are given by:

$$\phi = \cosh(kx) \sin(ky) + \cos(kx) \cosh(ky) \quad (1)$$

$$\psi = \sinh(kx) \sin(ky) - \sin(kx) \sinh(ky), \quad (2)$$

where k is a non-trivial root of the equation:

$$\tanh(kh) = -\tan(kh), \quad (3)$$

and ω , the sloshing frequency is given by:

$$\omega^2 = g k \tanh(kh). \quad (4)$$

Similarly, the asymmetric modes are given by:

$$\phi = \sinh(kx) \sin(ky) + \sin(kx) \sinh(ky) \quad (5)$$

$$\psi = \cosh(kx) \cos(ky) - \cos(kx) \cosh(ky), \quad (6)$$

where k is a non-trivial root of the equation:

$$\tanh(kh) = \tan(kh), \quad (7)$$

and ω is given by:

$$\omega^2 = g k \coth(kh). \quad (8)$$

The zero or trivial root of 3 and 7 correspond to the fundamental mode, where the velocity potential and streamfunction have an algebraic form:

$$\phi = xy \quad (9)$$

$$\psi = x^2 - y^2, \quad (10)$$

with ω given by:

$$\omega^2 = \frac{g}{h}. \quad (11)$$

Table 1 compares the absolute error between the analytical and numerical eigenvalues. Here the size of the matrix A is 1145×61 . As with the rectangle, the numerical eigenvalues in table 1 were obtained by excluding the Newman part in the basis functions, as it was found to increase the accuracy.

Table 1: In this table we compare the errors between the analytical and numerical sloshing frequencies for our isosceles triangular tank.

Sloshing Mode	Absolute Error
0	10^{-16}
1	10^{-12}
2	10^{-15}
3	10^{-11}
4	10^{-14}
5	10^{-9}
6	10^{-9}
7	10^{-5}
8	10^{-6}

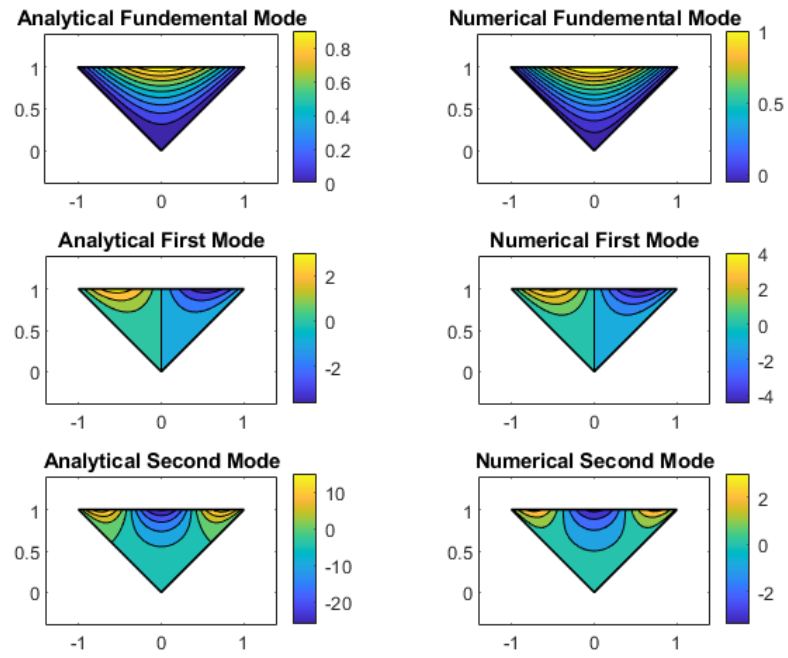


Figure 1: This figure compares the contour plots of the analytical and numerical streamfunction. Analogous to the rectangle, the accuracy of the modes tends to reduce significantly for higher modes.

References

- [1] G. Kirchhoff. “Ueber stehende Schwingungen einer schweren Flüssigkeit”. In: *Annalen der Physik* 246.5 (1880), pp. 34–46. DOI: <https://doi.org/10.1002/andp.18802460504>.
- [2] H. Lamb. *Hydrodynamics*. Dover Books on Physics. Dover publications, 1945. ISBN: 9780486602561.